

A model of Punitive Voting.*

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Abstract

I analyze a two-period model of political competition where voters care about candidates' integrity. Candidates must trade off implementing their preferred policy against maintaining their electoral promises. Voters punish candidates that deviate from electoral promises by voting for their opponent. I find that punitive voting can exert political discipline only if candidates face low levels uncertainty about voters preferences. In this case candidates' electoral promises are a compromise between their preferred policy and voters' preferences, and when elected they implement their promise. Finally, I show that when one candidate's ideal policy is closer to the median voter, an equilibrium exists where one candidate is disciplined and the other is not.

Keywords: Reputation, elections, political promises.

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1 Introduction

Our country is being run by incompetent people. And I won't be angry when we fix it, but until we fix it, I'm very, very angry. ¹

Donald J. Trump

Voters often cast their ballots out of frustration and disillusionment. [Wuthnow \(2018\)](#) documented that the political discontent of rural voters in America significantly fueled populist movements, including the support for Donald Trump in the 2016 election. He argues that many rural voters turned to Trump not necessarily because they endorsed his policies, but because he resonated with their anger and resentment towards a political establishment they felt had abandoned them. [Born, van Eck, and Johannesson \(2018\)](#) analyzed the impact of electoral promises on voter behavior in a controlled setting. They show that when politicians break their promises, voters retaliate by using their votes to punish them. Similarly [Galeotti and Zizzo \(2015\)](#) experimentally measures voters' reactions when candidates face a trade-off between competence and honesty, observing that voters tend to prioritize honesty, even when it leads to lower payoffs. These experiments suggests a strong voter bias towards integrity, reflecting deeper emotional and ethical concerns in electoral behavior. This pattern aligns with broader trends. As [Tambe \(2018\)](#) shows, in countries where voters lack trust in elections or party responses, voter turnout and participation tend to be low.

Voters are never pivotal in determining election outcomes, yet they are often motivated by underlying behavioral factors such as civic duty and psychological satisfaction, among others ([Blais, 2000](#)). However, one key incentive that has been largely overlooked is anger. While ubiquitous in media portrayals, social scientists have often ignored the impact of voter disenchantment and how perceived dishonesty can “flip” voters' decisions.

In this paper, I propose a new theory of electoral competition where voters' concern about dishonesty plays a central role, leading them to punish candidates who cannot be trusted. The model features two candidates, L and R , each with distinct policy preferences, and a voter positioned between them. Candidates are unsure about the exact position of the voter, but they share a common distribution over it. In the first period, candidates make campaign promises, and the voter chooses her preferred option. Once elected, a candidate may implement a policy that deviates from her promise, leading the voter to downgrade their perceived integrity. Although the voter has single-peaked preferences over policies, her preference for a candidate is influenced by the integrity associated with that candidate's promises. Consequently, a dishonest candidate who pledges to enact the voter's ideal policy may be less appealing than an honest candidate who promises a less ideal policy. The

¹As seen in ([Boylan, 2016](#)).

first challenge candidates face is deciding on a campaign promise that strikes a balance between aligning closely with the voter’s preferred policy—necessary to win elections—and staying true to their preferred policy, allowing them to act with integrity if elected. Once in office, incumbents must face the tension between the short-term temptation to pursue self-interest, which could jeopardize their future electoral success, and the long-term benefits of maintaining honesty to maximize their prospects in future elections. In the second period, another election takes place, during which candidates once again choose their promises and policies, and the voter updates his evaluation of each candidate’s integrity. Payoffs are determined by the policies enacted. In an extension of the model, I assume that the voter’s expected ideal policy is not equidistant between the two candidates’ bliss points, but instead closer to one of them. This proximity influences the more distant candidate’s likelihood of reneging on their promise once elected.

This paper demonstrates that concerns about punitive voting play a crucial role in shaping electoral dynamics and policy implementation. The main finding reveals a non-monotonic relationship between electoral promises and uncertainty regarding voter preferences in equilibrium. When there is an intermediate level of uncertainty about voter preferences and the voter is sufficiently sensitive to integrity, the threat of punitive voting serves as a disciplining mechanism. In this scenario, candidates are incentivized to craft promises that balance the voter’s ideal policy with their own preferences. Given their uncertainty about how voters will respond in equilibrium, candidates may be willing to risk a small chance of losing in exchange for making promises closer to their preferred policy. If these promises are close enough to their preferences—and assuming that candidates value future elections and voters prioritize integrity—the winning candidate in the first period will fulfill their promise. This leads to a second period where both candidates are perceived as having similar levels of integrity.

However, sustaining political integrity is challenging and only likely to emerge under the specific conditions outlined above. When these conditions are not met, candidates often fail to implement their campaign promises, with their decisions being heavily influenced by the level of uncertainty regarding voter preferences. When uncertainty is low, candidates understand the high risk of deviating from the voter’s ideal policy, which significantly increases their chances of losing the election, leading them to converge on similar promises. Yet, these promises are typically far from the candidates’ true preferences, prompting the eventual winner to act dishonestly after being elected, sacrificing future payoffs. Conversely, when uncertainty is high, dishonesty is encouraged in a different way. Due to the lack of precise information about the voter’s ideal point, the perceived importance of integrity in the second stage is overshadowed by the uncertainty surrounding the preferred policy, reducing the emphasis candidates place on honesty. As a result, acting dishonestly in the first stage does not significantly harm a candidate’s prospects in the final stage. Despite

this high uncertainty, candidates are still motivated to make promises that moderately align with both the voter's and their ideal policies to minimize the cost of dishonesty in the second period.

I also extend the analysis to asymmetric *bliss points*—candidates with different ideal policies—because real elections often feature unequally polarized opponents. Introducing asymmetry reshapes equilibrium behavior: the more distant candidate has stronger incentives to promise moderation and then renege in office, while the closer candidate can be disciplined to keep a modestly polarized promise. This asymmetry generates honesty–dishonesty configurations that symmetric models cannot capture and shows that the paper's main insights are robust even when candidates are not mirror images of each other, filling a gap in existing analyses.

Literature Review This paper contributes to the extensive literature studying electoral competition via promises. The first economic model of the political parties' behavior is presented in Downs (1957), where office-motivated candidates compete as rational agents to win election. In this paper, parties have an intrinsic value of gaining office and are not motivated by policies *per se*. On it, governments try to maximize the number of votes to stay in power as in Hotelling (1929) with consumers. This analysis misses the ideological motivation of parties. Barro (1973) presents a model of repeated election to study the behavior of elected public office-holders when they care both about reelection and policies. He states that candidates act selfishly choosing their preferred policy if there is no political control by citizens and apply the promised policies when the public control of candidates is present. In Barro (1973) voters are backwards-looking and lying is a factor that *discards* political representatives from being reelected when they act in their own interest.² The two previous models predict a fight for the median voter that might fail to explain some of the electoral promise location that is observed in reality. Like him, I assume retrospective voting, meaning voters evaluate incumbents based on whether they fulfilled past campaign promises. This is a departure from models with purely prospective voting (where voters only consider promises about the future). By building this into the model, I capture the empirical phenomenon of public criticism and backlash against promise-breakers. However, by allowing this cost to be linear in the deception, I predict that lies can happen in equilibrium. Wittman (1983) and Calvert (1985) introduce candidates that care both about the probability of winning and the policy that is applied. A similar result appears in this paper, where the joint role of integrity and uncertainty is enough to create polarization in the electoral promises. In this line, Alesina (1988) develops a model of forward-looking voters and ideologically motivated candidates without commitment. The absence of commitment creates an equilibrium of the one-shot game where each candidate promises her ideal pol-

²See also Austen-Smith and Banks (1989).

icy due to a problem of credibility. Convergence only appears when the game is infinitely repeated. My setup allows for both types of equilibria depending on the importance of the reputation and the level of uncertainty within the model.

The literature on electoral competition without commitment is directly relevant. In Banks (1990), credibility is restored by an *explicit deviation cost* borne by the winner within a sequential-equilibrium framework; Callander and Wilkie (2007) similarly discipline misrepresentation via a lying cost borne by candidates. By contrast, I place discipline on the voter side: deviations generate a *retrospective electoral penalty*. This approach does not require intrinsically honest politicians—discipline arises from voter punishment—and, holding platforms fixed, it implies reelection probabilities decline with the realized promise–policy gap and feature a sharp cutoff once the gap is large enough. Fournier and Francou (2023) study location games with reference policies and losses from deviating relative to that reference. This maps neatly to my period-0 *implementation* subgame: conditional on winning, the incumbent chooses a policy relative to the reference set by her promise. The difference is that, in my environment, the loss is translated into a period-1 *retrospective electoral penalty*. My contribution is to endogenize the reference via strategic promises and to link deviation losses to reelection through voter punishment under aggregate uncertainty. A related debate concerns the *shape* of the deviation cost. Much of the credibility literature adopts a *linear* or piecewise-linear loss for tractability and transparent trade-offs (e.g., Banks (1990); Callander and Wilkie (2007)), while other work and evidence suggest costs that rise with the size or salience of the breach (convex) or display threshold responses; reference-dependent formulations such as Fournier and Francou (2023) accommodate these possibilities. In the appendix I consider all three specifications—linear, concave, and convex—and show which predictions survive: under concave costs the three-region structure and promise patterns remain with shifted cutoffs and a bang–bang policy choice; under convex costs the honest region disappears, the winner shades by an interior amount, and promises become more polarized to cap the future promise–policy gap, with some interior objects characterized implicitly rather than in closed form. With period 1 data alone, my retrospective penalty is observationally equivalent to a candidate-side deviation or lying cost as in Banks (1990) and Callander and Wilkie (2007), or to a reference-dependent loss as in Fournier and Francou (2023): holding platforms fixed, reelection probabilities decline linearly in the realized promise–policy gap.

Aragonés, Palfrey, and Postlewaite (2007) study how reputation is used by voters to form beliefs about the credibility of candidates’ campaign promises. They show that there is an equilibrium where candidates announce policies different from their ideal point and keep their promises. The voter I present in this paper is, instead backward-looking. In my model, candidates incur reputation costs when they apply a policy that differs from their promises. This cost harms the predisposition of citizens to vote for the candidates that they

consider reliable as a sort of punishment.³

Weelden (2013) analyzes the trade-off between applying rent-seeking policies or increasing the probabilities of reelection with an infinite number of candidates. He extends the analysis of credibility of campaign promises when there is no full commitment. In the same line, Bischoff and Siemers (2013) explain how retrospective voting and biased beliefs affect the policy outcomes and the probabilities of reelection. List and Sturm (2006) analyze the use of policy promises in campaign in a two-dimension policy space. They find that there exist strong effects to approach to the voter in secondary policy issues. The relation between the promises and the policy intentions of candidates is also studied in Schnakenberg (2016). He proves that cheap talk can also be informative even if there is no commitment. In this line, Kartik and Van Weelden (2018) prove that cheap talk during the elections can affect candidates' behavior when they are in power. My model, is different from the *cheap-talk* model, as there is an endogenous *cost of lying* when candidates promise things that are different from the policies that they apply. Kartik and McAfee (2007) (through character) and Callander and Wilkie (2007) (through the cost of lying) develop models where candidates differ in their willingness to lie in elections and voters interpret promises in a signaling game to disentangle the different types.

Andreottola (2020) studies the incentives of a politician to hide information in order to not be considered as a *flip-flopper*. In his model, the optimal policy might change given new private information. Differently from my paper, he focuses on the information updating of an incumbent politician and how a strong reputation concern affects the application of the optimal policies, while I focus on the campaign polarization and the capacity of the repeated election to force honest behaviors. Rivas (2015) develops a model where a politician has to take several decisions during his term office. In his model, citizens are backwards-looking when evaluating the quality of the politician. Under this assumptions, he sees incentives to take selfish decisions first and act honestly later. This differs with my papers because he considers multiple decisions in one electoral period, so there is no threaten of no-reelection.

Differently from previous papers on re-election, I model reputation as an additive term that decreases the utility of the voter to choose certain candidate. This way of modeling is close to the literature of electoral competition and *valence*.⁴ Ashworth and Bueno de Mesquita (2009) develop a model where candidates can invest in costly valences and show that valences are more important the less polarized is the society. In their model, a shock in one candidate's valence leads to complete platform convergence, while in mine, reputation can keep polarization. Gouret, Hollard, and Rossignol (2011) analyze in a survey prior to

³For a literature review on retrospective voting see Healy and Malhotra (2013).

⁴See Stokes (1963), Ansolabehere and Snyder (2000) Groseclose (2001), Aragonès and Palfrey (2002), Hummel (2010), Aragonès and Xefteris (2017). Xefteris (2013), Denter (2021), and Buisseret and Van Weelden (2022) among others.

the French presidential election of 2007 several spatial voting models with valence. Taking into account the empirical evidence, they develop a model of intensity valence that diverges from the additive valence models. A related strand of work emphasizes the role of consistency costs in shaping electoral outcomes. In Aragonès and Xefteris (2017), the consistency penalty reflects the distance between a candidate’s chosen position and the party’s reference policy, so the cost is anchored in party identity. In contrast, my framework lets the politician choose both the campaign promise and the implemented policy. This grounds the penalty as a purely candidate-specific integrity cost that directly creates incentives to either lie or remain honest, thereby shaping both promises and policies. Similarly, Hummel (2010) models the consistency cost arising from differences between primary and general election platforms, so valence is tied to campaign-period positions. In my setting, however, voters evaluate politicians retrospectively and impose penalties based on the discrepancy between campaign promises and the policies enacted once in office, turning consistency into a mechanism of accountability across governing periods rather than only within electoral campaigns.

The central mechanism in this paper is the tension between winning with attractive promises and governing honestly afterward. Earlier treatments often rule out promise-breaking; I instead build it in and show how retrospective punishment reshapes behavior. This delivers a more realistic account of voter and candidate strategies and produces conclusions that contrast with—and improve on—standard predictions. The framework explains why moderate platforms sometimes fail to be implemented, why extreme policies persist, and how truth and lies can coexist on path. These features make the model a useful benchmark for thinking about accountability when promises are non-binding.

The rest of the paper is organized as follows: Section 2 presents the model, including the players and the timing. Section 3 develops the equilibrium of the model and, discusses the results and their implications. All the proofs are relegated to the Appendix B.

2 Model

In this section, I present the main features of the model: players, their payoffs, and the dynamics of information in the game.

Two *candidates* compete in an election to decide who will apply a policy. I define a policy as a real number, $x \in \mathbb{R}$. The timing of the election is the following: first, each candidate simultaneously announces a promise $p_i \in \mathbb{R}$ about the policy she will implement.

The voter observes these promises and votes for his preferred candidate; the candidate who wins the vote is elected and applies a policy, $\pi \in \mathbb{R}$. Then, the voter observes the policy implemented and computes the cost of lying to the winning candidate. Once this process finishes, it repeats itself. I analyze two periods: $t = 0$ and $t = 1$.

There are three players: two candidates, L and R (she), and a voter (he). I use i to refer to a generic candidate and v to the voter.

The voter Assuming the existence of just one voter is a simplification without loss of generality that allows me to focus on the behavior of candidates. It is equivalent to take any distribution of voters and consider that the winner is elected by majority voting. In this case, the analysis focuses on the actions of the median voter.

The voter will choose the candidate that he thinks is closer to him. For simplicity, I will not consider the possibility of abstention. Therefore, he must vote for one and only one candidate. He has an innate preference over the policy, that I normalize to zero, and it is common knowledge to all players. The voter decides his vote without thinking about future elections; he will punish those candidates who lie. I assume this as my main goal is to study how the promises are shaped in the presence of reelection. I want to capture the punishment effect that deviations from the promises made in the campaign provoke; this is why I don't allow the voter to make any inference about the preferred policy of candidates.

Cost of lying The *cost of lying* is the key factor in this model. I will refer to it as cost of lying, integrity cost, or reputation cost. It measures the difference between a candidate's promises and what she effectively does. The voter calculates the cost of lying of each candidate at every period, which will affect the chances of election. By assumption, this term at $t = 0$ will be zero for both candidates.

Equation (1) describes the cost of lying of candidate i at the beginning of period 1.

$$\mathfrak{C}_i^1 = \delta |p_i^0 - \pi_i^0| \quad (1)$$

where $\delta \in \mathbb{R}^+$.⁵ Equation (1) states that the reputation cost of a candidate consists of the positive difference between the promise made during election and the policy applied after

⁵By design, deviations from promises are intentional: there are no implementation constraints or unforeseen shocks in office. The voter punishes realized gaps $|p - \pi|$ irrespective of their sign. In equilibrium, deviations occur in the direction of the winner's bliss point, i.e. away from the voter's ideal policy, so this reduced form does not overturn any comparative statics.

winning. The interpretation of the term $\mathfrak{C}_i^1$ is the following: it increases when a candidate deviates from her promise. δ captures the effect of the difference between promises and policies on voter's decisions. The term $\mathfrak{C}$ distinguishes this model from a cheap-talk model, as it is an endogenous cost of lying that relates promises and policies through the probability of winning.

Notice that in period one, the reputation cost enters in the utility cost of the voter as a valence shift. For tractability I assume linearity in (1). In Appendix A, I generalize to $\mathfrak{C}_i^1 = \delta |p_i^0 - \pi_i^0|^\kappa$ with $\kappa > 0$. This allows me to discuss the change in equilibria as a function of the concavity/convexity of the cost function. Proofs and comparative statics are provided in Appendix A.

Candidates do not know exactly the preferred candidate for the voter given $\mathbf{p}$ (the full vector of promises) and $\boldsymbol{\pi}$ (the full vector of policies); I assume that the voter has a private preference for one of the candidates that affects his willingness to vote for her but does not depend on $\mathbf{p}_i$ or $\boldsymbol{\pi}_i$. This preference is a pair of random variables A^0, A^1 *i.i.d.* that follow a uniform distribution in the interval $[-a, a]$, where $a \geq 0$ and affects the utility for the voter of choosing candidate R . The utility of the voter at the period t is given by the next equation.

$$\mathcal{U}_v^t(p_i^t, \mathfrak{C}_i^t) = \begin{cases} -|p_L^t| - \mathfrak{C}_L^t & \text{if he votes for candidate } L \\ -|p_R^t| - \mathfrak{C}_R^t - A^t & \text{if he votes for candidate } R \end{cases} \quad (2)$$

where p_i^t are the promises of candidate i at period t (see below). As (2) shows, the voter receives utility from each candidate considering two factors. The first factor is the distance from the promises of each candidate to her preferred policy, 0. The second is this candidate's cost of lying. An increase in $\mathfrak{C}_i^t$ decreases the utility of choosing candidate i for the voter. Hence, given a pair of promises, it will diminish the candidate's chances of being elected. As there are two candidates, there is always at least one maximizing the utility of voter v . The probability of candidate R winning is equivalent to $P(A^t \leq -|p_R^t| + |p_L^t| + \mathfrak{C}^t)$. Using the fact that A^t is uniformly distributed, the probability of candidate R winning reads as,

$$P(\text{Win } R) = \begin{cases} 0 & \text{if } -a \geq -|p_R^t| + |p_L^t| + \mathfrak{C}^t \\ \frac{1}{2a} (-|p_R^t| + |p_L^t| + \mathfrak{C}^t + a) & \text{if } -a \leq -|p_R^t| + |p_L^t| + \mathfrak{C}^t \leq a \\ 1 & \text{if } a \leq -|p_R^t| + |p_L^t| + \mathfrak{C}^t \end{cases}$$

where $\mathfrak{C}^t = \mathfrak{C}_L^t - \mathfrak{C}_R^t$. The cost of lying makes promises *costly*, because they endogenously reduce a candidate's probability of election. Candidates are punished by the voter if they

move away from their preferred policy; however, a candidate can still win with positive probability even if she does not promise exactly what the voter wants. *The voter is mechanical: given promises, implemented policies, and the taste shock, a fixed voting operator maps observables into the winner, so the equilibrium is defined solely by the candidates' strategies and the probability of winning. Finally, the random variable that affects the voter's utility is realized once per period and the two realizations are i.i.d.; this lets me focus on the role of uncertainty in politicians' choices and ignore information updating that would arise under cross-period correlation.*

Candidates Each candidate has an exogenously given preferred policy,⁶ x_R and $x_L \in \mathbb{R}$ for candidates R and L , respectively. These points are assumed to be symmetric around zero. Without loss of generality I assume $x_R = 1$ and $x_L = -1$. At the beginning of the period, each candidate publicly announces p_i^t , which is a promise about the policy that the candidate will implement if elected. The winner of the election decides π_i^t . For notation simplicity, I set $\pi_i^t = p_i^t$ for the losing candidate, as she does not face a cost of lying. I say that a candidate is dishonest whenever she chooses $p_i^t \neq \pi_i^t$.

To simplify notation, I will use through the paper $\mathfrak{C}^t$ to refer to the integrity cost for both candidates, $\mathbf{p}^t$ for the promises, and $\boldsymbol{\pi}^t$ for the policies at time t . I will remove the superscript t when it refers to the complete history.

Flow utility Candidates act as rational agents, maximizing their utility. The utility function of candidate i for period t is as follows:

$$u_i^t(\pi_i^t; \pi_{-i}^t, \mathbf{p}^t, \mathfrak{R}^t) = \begin{cases} -|x_i - \pi_i^t| & \text{if } i \text{ wins} \\ -|x_i - \pi_{-i}^t| & \text{if } i \text{ loses} \end{cases} \quad (3)$$

Where π_{-i}^t is the policy applied by the other candidate when she wins. The utility that candidates perceive from winning election comes from the opportunity to implement a policy. For this reason, despite the candidates are not office motivated, they seek to win. With this, candidates maximize their total utility for the game.

$$U_i(\mathbf{p}, \boldsymbol{\pi}) = \sum_{t=0}^1 \gamma^t u_i^t(\pi_i^t; x_i, \pi_{-i}^t) \quad (4)$$

⁶The ideal point of each candidate can be also interpreted not as a preferred policy, but as the policy that they think is ideal to everyone.

for $0 < \gamma < 1$ a time discount factor. The utility of the game for candidates is the discounted sum of utilities for each period.

To sum up, candidates must choose every period a promise, p_i^t , and a policy to implement, π_i^t . Hence, the strategy of the candidate i at time t is a pair $\{p_i^t, \pi_i^t\}$. Consequently, the candidate's strategy for the game will consist of a sequence of promises and actions $\{p_i^t, \pi_i^t\}_{t=0}^1$. The structure of one period is as follows:

1. First, on *Campaign*, candidates publicly announce their policies, interpreted as intentions about policy implementations, p_i .
2. Second, in the *Voting stage*, the voter decides which candidate gives him a higher utility and votes.
3. Finally, in the *Office stage*, the elected candidate chooses a policy π_i . The voter observes it and updates candidate's cost of lying.

3 Analysis

In this section, I develop all the necessary steps to solve the model and present the equilibrium of the game. The equilibrium is a subgame-perfect Nash equilibrium for the candidates' strategies. It is a pair of promises and policies for each individual and each period:

$$\{p_R^0, p_R^1, \pi_R^0, \pi_R^1\}, \{p_L^0, p_L^1, \pi_L^0, \pi_L^1\}$$

Because the voter is mechanical and the only uncertainty is an i.i.d. taste shock $A_t \sim U[-a, a]$ (independent of histories and actions), beliefs are uniquely pinned down by the uniform prior and never discipline behavior. I therefore analyze the reduced game in expected utilities and solve for a subgame perfect equilibrium by backward induction. Compact action sets and continuity ensure best replies exist in every subgame. The resulting subgame perfect outcome coincides with the Perfect Bayesian and sequential equilibria of the original extensive form with chance moves.

In Banks (1990), credibility operates through private information and belief updating over candidate types, and off path beliefs are payoff relevant. Sequential equilibrium is therefore required to specify beliefs at unreached information sets and to ensure sequential rationality. By contrast, my environment has no private types, the only randomness is an i.i.d. taste shock that I integrate out, and the voting rule is mechanical. As a result, solving

the expected payoffs game by backward induction delivers an SPE that coincides with the Bayesian solution, while discipline comes from a voter imposed retrospective penalty rather than an intrinsic deviation cost borne directly by the winner.

The next set of results characterizes the equilibrium strategies in the last period, $t = 1$. Later, I will solve the first period.

3.1 Last period

In the last period, each candidate chooses a promise and a policy. This decision determines candidates' integrity cost, $\mathfrak{C}_i^1$. For this reason, the decision on p_i^1 and π_i^1 are not motivated by their effect on future lying costs as will happen in the first period. Instead, when candidates decide on a promise and a policy to implement, they only care about the impact on that period's utility. Proposition 3.1 states the policy that candidates apply in equilibrium using this reasoning. As there is no mechanism that the voter can use to punish her, the candidate that wins election in the last period maximizes the flow utility of $t = 1$.

Proposition 3.1 (Optimal policies for the last period). *In the last period, the candidate that wins the election best replies with her preferred policy in equilibrium:*

$$\pi_i^1 = x_i.$$

The best reply of the previous proposition holds for every promise, p_i^1 , and integrity cost, $\mathfrak{C}_i^1$, given that she has won the election. As candidate i is the winner, there is no possible punishment from the voter. Therefore, she chooses in equilibrium the policy that maximizes his utility. Once a candidate is elected to choose the policy, the probability of winning election does not play a role in her decisions. As the policy in the last stage does not depend on the promise that made the candidate win, she will apply her preferred policy. This is the only result independent on the variable A^t .

Using the previous result for the optimal policies for the last period, I can state the equilibrium payoffs for each candidate conditional on winning and losing. If candidate i wins the final period's election, she will apply her preferred policy and get a utility given by:

$$u_i(\pi_i^1; p_i^1, p_{-i}^1, \pi_{-i}^1) = -|x_i - \pi_i^1| = -|x_i - x_i| = 0. \quad (5)$$

If instead, candidate i loses in the last period, the other candidate will apply her preferred policy, and then the utility for losing is

$$u_i(\pi_i^1; p_i^1, p_{-i}^1, \pi_{-i}^1) = -|x_i - \pi_{-i}^1| = -|x_i - x_{-i}| = -2. \quad (6)$$

This result allows me to find the promises in equilibrium, as the optimal policies determine the flow utilities of both losing and winning. If candidates arrive in period one with the same $\mathfrak{C}_i$, they must choose the voter's preferred policy in order to win, as the voter can only differentiate between them by their promises. This is not the case if they have different costs of lying. If, for example, candidate L arrives at the last period with big lying costs, her opponent has a *relative advantage* that she can exploit to win that period's election. In this case, candidate L can not do anything to win, as candidate R will always win in equilibrium promising the voter's ideal policy. As only one candidate wins the election at each period and both begin the game with zero integrity cost, in equilibrium, there is only one case in which both candidates have the same reputation in the last period. This is if $\mathfrak{C}_L^1 = \mathfrak{C}_R^1 = 0$. Proposition 3.2 states the optimal policy promises for both candidates in the last period of the game depending on the integrity costs.

Proposition 3.2 (Optimal promise in the last period). *If $\mathfrak{C}_L^1 \geq a$ there exist multiple equilibria of the form*

$$p_R^1 \in [a - \mathfrak{C}_L^1, a + \mathfrak{C}_L^1].$$

If $a > \mathfrak{C}_L^1 \geq 0$, $\mathfrak{C}_R^1 = 0$, the unique equilibrium is

$$p_R = p_L = 0.$$

In the final election, promises are just vote-getting messages. If the opponent's integrity hit from period 0 is bigger than the uncertainty facing voters, a candidate can always secure victory by promising close enough to the median (there's a small interval around zero that guarantees a win). If not, both candidates promise the median voter's ideal policy. With a positive, but small enough, integrity cost both candidates have a positive probability of winning in equilibrium and they still promise the ideal point of the voter. This enlarges the strategies that might be optimal in the first period with respect a case where there is full information, as incurring in integrity costs is not a disqualifier. Next section characterizes the results for the first period of the game and states the equilibria of the game.

3.2 First period

Given the results of the last period of the game, I can calculate the equilibrium of the game. The structure of this section is similar to the previous one. First, I will characterize the optimal policies for any possible promise. Second, I will use this result to find the promises of both candidates in equilibrium. One particularity of the first period is that, by assumption, both candidates are symmetric for the voter's ideal point; they have a zero integrity cost, and their preferred policy is symmetric around the voter.

The candidate that wins the election must choose which policy to apply. This decision can make her keep her initial integrity or incur in a cost by lying. This decision is one of the keys to the model. The policy that the candidate applies determines her level of integrity for the next period and consequently her chances to win. At the same time, the policies that the candidate applies in the first period depend on the promise that made her win. If this promise is close enough to the ideal point of the candidate, she will keep her promise and she will choose her ideal point otherwise. However, there is an extra condition for both candidates to keep their promise as Proposition 3.3 shows; the level of uncertainty about voter's preferences, a , must be smaller than the future discounted salience of the integrity cost, $\gamma\delta$.⁷

Proposition 3.3 (Optimal policies in the first period). *If candidate R wins at $t = 0$ with p_R^0 , she chooses, in equilibrium,*

$$\pi_R^0 = \begin{cases} p_R^0 & \text{if } a < \gamma\delta \text{ and } 1 + \gamma \geq p_R^0 \geq 1 - \gamma, \\ 1 & \text{Otherwise.} \end{cases}$$

In period 0, the winner either keeps the promise or reverts to her bliss point—nothing in between. She keeps the promise only when uncertainty is low relative to the future salience of integrity, $a < \gamma\delta$, and the promise isn't too far from her bliss point. The optimal policy for candidate L is symmetric to Proposition 3.3. The optimal policies in the first period are only the ideal point or keeping the promise. As the utility function is linear, there is no possibility of finding an optimal promise in the middle between these two options. Assuming a quadratic utility for the candidates, for the voter, or for both can make different equilibria appear where $\pi_R^0 \in (p_R^0, 1)$.

Before stating the optimal promises in equilibrium for the first period, it is important to note that, at the beginning of the game, the expected utility of candidates can have three different shapes. These shapes depend on the relation between the parameters of the model

⁷As both A^t follow a uniform distribution, a is a measure of the variance of A^t .

and the promise p_R^0 . If the candidate is honest in the first period, $\pi_R^0 = p_R^0$, the expected utility of candidate R reads as

$$\mathbb{E}[U_R(\mathbf{p}, \boldsymbol{\pi})] = -\frac{1}{2a} (-|p_R^0| + |p_L^0| + a) [1 - p_R^0 + \gamma] - \frac{1}{2a} (a + |p_R^0| - |p_L^0|) [1 - p_L^0 + \gamma],$$

where $\mathbf{p}$ is the vector of promises and $\boldsymbol{\pi}$ the vector of policies. The first term corresponds to the probability of winning in the first period multiplied by the utility of winning and applying the promise $-|1 - p_R^0|$ plus the expected utility of the last period, that is -2 with probability one half. As both candidates have the same level of integrity in the last stage, both have the same chances of winning in equilibrium. The second term is the probability of losing times the utility of candidate L applying her policy.

If the candidate applies a different policy to her promise, the utility function for candidate R is

$$\mathbb{E}[U_R(\mathbf{p}, \boldsymbol{\pi})] = \begin{cases} -\frac{1}{2a} (-|p_R^0| + |p_L^0| + a) \left[\frac{\gamma}{a} (\delta|1 - p_R^0| + a) \right] \\ -\frac{1}{2a} (a + |p_R^0| - |p_L^0|) \left[2 + \frac{\gamma}{a} (-\delta|1 - p_L^0| + a) \right] & \text{if } \delta|1 - p_R^0| < a, \\ -\frac{1}{2a} (-|p_R^0| + |p_L^0| + a) 2\gamma \\ -\frac{1}{2a} (a + |p_R^0| - |p_L^0|) 2 & \text{if } \delta|1 - p_R^0| \geq a. \end{cases}$$

Notice two things. First, the fact that the separation between both expressions depend on the relation between $\delta|1 - p_R^0|$ and a is because of the optimal promises in the last stage. Second, this is only true in the case of symmetry.

The first term of both expressions correspond with the probability of winning multiplied by the utility of winning. The first period utility is zero, as candidate applies her ideal point. The only difference is the probability in equilibrium of winning in the second stage. If $\delta|1 - p_R^0| < a$ this probability is positive. In the opposite case, this probability is zero and the candidate loses with certainty. Notice that the expected utility is a continuous function in the three cases. If $\delta|1 - p_R^0| = a$, the two previous expressions are equal. The same holds for $p_R^0 = \pi_R^0$. The next proposition characterizes the optimal promises in the first period as a function of the relation between a and $\gamma\delta$.

Proposition 3.4 (Optimal promises for the first period). *The right candidate's equilibrium*

promise p_R^0 is

$$p_R^0 = \begin{cases} \frac{a}{2} & \text{if } 2(1 - \gamma) \leq a \leq \gamma\delta, \\ 1 + \frac{a}{2} - \frac{a}{\gamma\delta} & \text{if } a > \gamma\delta, \ 2(1 - \gamma) < \gamma\delta, \ a \leq \frac{2\gamma\delta}{2 - \gamma\delta}, \\ 0 & \text{otherwise.} \end{cases}$$

By symmetry, the left candidate's promise is $p_L^0 = -p_R^0$.

There are three promise regimes at $t = 0$, partitioned by the thresholds $2(1 - \gamma)$, $\gamma\delta$, and $\frac{2\gamma\delta}{2 - \gamma\delta}$.

The first regime is the *honest-polarized region under intermediate uncertainty*. If $2(1 - \gamma) \leq a \leq \gamma\delta$, each candidate announces a platform that moves linearly away from the voter as uncertainty rises. For R , $p_R^0 = a/2$ (and symmetrically $p_L^0 = -a/2$). The interval $[2(1 - \gamma), \gamma\delta]$ expands with the importance of the future, so higher γ or higher δ makes this region larger; it is nonempty only if $\gamma > \frac{2}{3}$. In this region the winner *keeps* the promise in office.

The second is a *high-uncertainty interior-dishonest region*. If $a > \gamma\delta$, $2(1 - \gamma) < \gamma\delta$, and $a \leq \frac{2\gamma\delta}{2 - \gamma\delta}$, promises move *toward* each candidate's bliss point as uncertainty rises. For R , $p_R^0 = 1 + \frac{a}{2} - \frac{a}{\gamma\delta}$ (and $p_L^0 = -1 - \frac{a}{2} + \frac{a}{\gamma\delta}$). Promises lie weakly between the voter's ideal and each candidate's bliss point, and as a increases they converge toward the bliss points. When $\gamma\delta$ approaches a from above, this interior regime connects to the honest-polarized one; when a exceeds the upper bound or $\gamma\delta \leq 2(1 - \gamma)$, it disappears. In this region the winner *reneges* in office.

The third offers *central promises* for all the other parameter cases. In all remaining parameter values, both candidates promise the voter's ideal policy, $p_R^0 = p_L^0 = 0$. This occurs either because uncertainty is too low for polarization to pay when the future matters a lot, or because the future salience of integrity is too small relative to the other parameters to sustain either polarized or interior promises.

The next theorem presents all three different types of equilibria that exist in this model. Notice that equilibria in the second stage depend on the relation between reputation and uncertainty, as stated at the beginning of this section. If the cost of lying of one candidate exceeds the value of the uncertainty, the other candidate has a margin of movements on her promises around the ideal policy of the median voter that allows him to win with certainty in equilibrium. In the opposite case, both candidates will promise the voter's ideal policy.

Theorem 3.1 (Equilibria of the game). There are three equilibrium patterns, partitioned by the thresholds $2(1 - \gamma)$, $\gamma\delta$, and $\frac{2\gamma\delta}{2 - \gamma\delta}$.

1. *Honest equilibrium.* If $2(1 - \gamma) < a \leq \gamma\delta$, in the first stage both candidates choose polarized promises that move linearly with a :

$$p_R^0 = \frac{a}{2}, \quad p_L^0 = -\frac{a}{2}.$$

The winner *keeps* her promise in office, $\pi_i^0 = p_i^0$. In the second stage both promise the voter's ideal policy (0), and the winner implements her own ideal policy.

2. *High-uncertainty interior-dishonest equilibrium.* If $\gamma\delta < a \leq \frac{2\gamma\delta}{2-\gamma\delta}$ and $2(1 - \gamma) < \gamma\delta$, in the first stage both candidates choose interior promises that shade toward their bliss points:

$$p_R^0 = 1 + \frac{a}{2} - \frac{a}{\gamma\delta}, \quad p_L^0 = -1 - \frac{a}{2} + \frac{a}{\gamma\delta}.$$

The winner *reneges* and implements her ideal policy in period 0. In the second stage both promise 0, and the winner implements her ideal policy.

3. *Central-promise equilibrium (residual cases).* In all remaining parameter values, in the first stage both candidates promise the voter's ideal policy,

$$p_R^0 = p_L^0 = 0,$$

and the winner implements her ideal policy. In the second stage:

- if $\delta > a$, the candidate with zero lying cost (from period 0) chooses a promise in $[a - \delta, \delta - a]$, wins with probability one, and implements her ideal policy;
- if $\delta \leq a$, both promise 0 and the winner implements her ideal policy.

To understand this theorem, I consider the different cases based on the future discount and the reputation salience. When $2(1 - \gamma) > \gamma\delta$, the optimal first-stage promises for both candidates are the voter's ideal policy, independently of a (the level of uncertainty). Note that if $\gamma < \frac{2}{3}$ this inequality always holds. However, whoever wins still implements her own ideal policy in office. Integrity effects also matter in the second stage: after the first election, either one candidate has zero probability of winning (when $\delta > a$) or both still have positive probabilities (when $\delta \leq a$).

If both candidates retain a positive chance of winning in the second stage, their optimal second-stage promises are the voter's ideal policy. Here, uncertainty about the median voter's decision is high relative to the salience of integrity: the exogenous taste shock dominates the endogenous integrity term. Any small deviation from the voter's ideal is second-order relative to that uncertainty, so both candidates continue to compete for the center in equilibrium.

If instead one candidate's lying cost is large enough that her second-stage win probability collapses to zero (when $\delta > a$), the second-stage interaction is equivalent to a full-information

race; I refer to this as the *complete information equilibrium*. The candidate who lost the first election can choose a promise in a neighborhood of the voter's ideal that guarantees victory in the second stage. I can select a particular point in that interval by assuming an infinitesimal *legacy effect* that nudges candidates toward honesty even after the game is effectively decided; this pins down $p_R^1 = \delta - a$ and $p_L^1 = a - \delta$. In consequence, incurring an integrity cost in period 0 leads the advantaged candidate to choose a more polarized second-stage promise: given her advantage, she can win with certainty while revealing the policy she will implement.

Now consider the two equilibria that arise when $2(1-\gamma) < \gamma\delta$. If uncertainty lies between these two terms, candidates are willing to polarize their first-stage promises away from the voter's ideal. Because the future weight of integrity is high relative to uncertainty, they accept some loss in immediate win probability in exchange for making a promise they will keep if elected. Only in this region do winners implement their promises in office. The logic is twofold: with low uncertainty, large promise–policy gaps would ensure a second-stage loss, so candidates prefer to keep the promise; but this also requires that the promise not be too far from the candidate's bliss point (otherwise the candidate would sacrifice future integrity to gain current policy utility). Theorem 3.1 implies that with arbitrarily small uncertainty candidates go honest when $\gamma = 1$; being truthful in period 0, both then enter period 1 with zero integrity costs.

If uncertainty is higher, $a > \gamma\delta > 2(1-\gamma)$, but still below $\frac{2\gamma\delta}{2-\gamma\delta}$, a different equilibrium emerges: the *high-uncertainty equilibrium*. First-stage promises now move *toward* each candidate's bliss point as a increases. Two forces are at work. On the one hand, the winner will not keep the period-0 promise, generating an integrity cost that lowers the second-stage win probability. On the other hand, high uncertainty makes the election outcome—and thus the bite of integrity—less predictable. Together, these forces push promises away from the voter's ideal: candidates accept lower first-stage win chances to reduce future lying costs and improve second-stage prospects. Once uncertainty exceeds $\frac{2\gamma\delta}{2-\gamma\delta}$, however, ignorance about voter behavior overwhelms integrity considerations, and candidates return to promising the voter's ideal policy.

3.3 Descriptive statics

Figure 1 shows the evolution of the promises in the first stage as a function of the level of uncertainty. First of all, note that promises are continuous when $\gamma\delta = a$, as

$$\lim_{a \rightarrow \gamma\delta} 1 + \frac{a}{2} - \frac{a}{\gamma\delta} = \frac{a}{2}$$

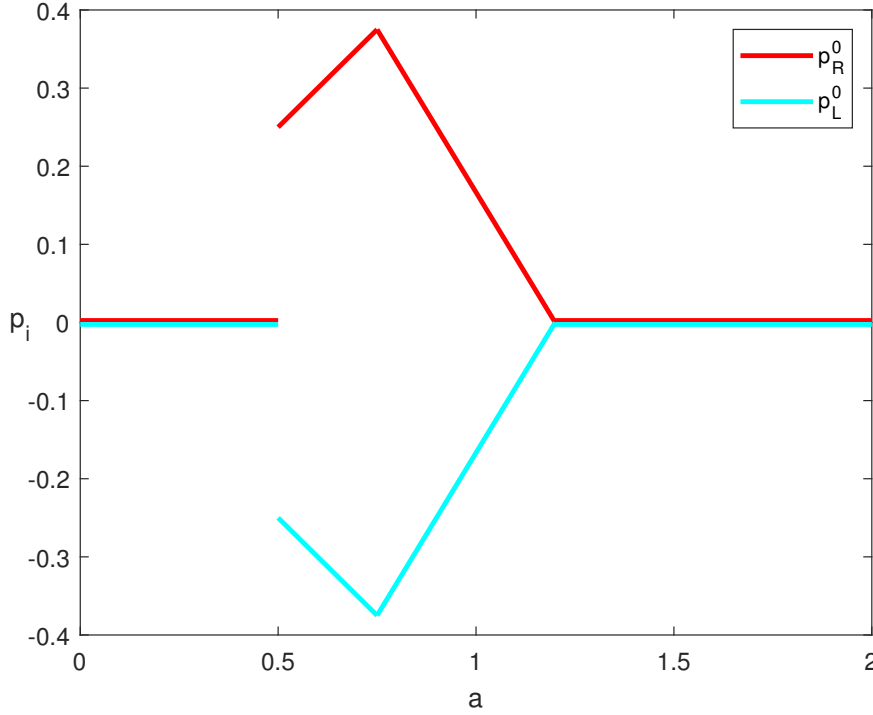


Figure 1: The promises in the first stage as a function of a . Parameters: $\gamma = 0.75, \delta = 1$.

and $\gamma\delta > \frac{2\gamma\delta}{2+\gamma\delta}$. Therefore, for small levels of uncertainty, $a < 2(1 - \gamma)$ candidates choose still the ideal policy of the voter as their promise corresponding with the *complete information equilibrium*. However, when the level of uncertainty increases, they move towards more polarized positions. If the level of uncertainty about voter's preferences continues increasing $a > \gamma\delta$, the effect is the opposite, as candidates chose more moderate promises. This effect continues until the level of uncertainty is high enough that candidates do not want to risk probabilities of winning in exchange of less integrity costs and go back to promise the voter's ideal policy. In consequence, being closer to the voter in the first stage (via promises) makes candidates go further in the second stage when they apply their ideal policy (via integrity).

Hence, integrity plays a double role. In the second stage, if candidates have kept a relatively small lying cost, they can still win with some positive probability. Knowing this, candidates do forward looking reasoning and introduce this effect on their decision-making. Therefore, in the first stage, the integrity effect can polarize candidates in their promises and not choose only the ideal policy for the voter. From the point of view of *honesty*, how promises and policies relate, this polarization can be positive, as it produces a lower deviation between them.

The thresholds in Theorem 3.1 (e.g. $a = \gamma\delta$, $a = 2(1 - \gamma)$) arise from the interaction of two features: (i) the absolute-value form of the integrity cost $|p - \pi|$, which introduces kinks into candidates' objective functions, and (ii) the uniform distribution of the voter's taste shock, which produces cutoff probabilities around a . Away from these thresholds, best-

response correspondences are continuous in parameters. At the cutoff $a = \gamma\delta$, promises are continuous, but the implemented policy need not be. For $a < \gamma\delta$ and $p \geq 1 - \gamma$, the winner keeps the promise ($\pi = p$). In this case, this clearly means that the policy is also continuous. For $a > \gamma\delta$ or $p \leq 1 - \gamma$, she reverts to the bliss point ($\pi = 1$). Exactly at $a = \gamma\delta$, the period 1 term is flat in π , so any $\pi \in [p, 1]$ is optimal and the implementation correspondence is $[p, 1]$. Consequently, the probability of keeping the promise typically *jumps* from one to zero at $a = \gamma\delta$.

Away from the thresholds $a \in \{2(1 - \gamma), \gamma\delta, 2\gamma\delta/(2 - \gamma\delta)\}$, payoffs are continuous and piecewise affine and the feasible sets are compact, so by the Maximum Theorem best replies are upper hemicontinuous in θ and the induced equilibrium promise correspondence is nonempty, compact-valued, and upper hemicontinuous on that domain.

As uncertainty a rises, the *marginal electoral return* to moving p toward the voter falls proportionally, while the *marginal policy cost* of moving away from the candidate's bliss point is unchanged. Moreover, whenever the anticipated reputation penalty satisfies $\delta|p - \pi| \geq a$, the period 1 win probability for the deviator collapses to zero (saturation). At the threshold $a = 2(1 - \gamma)$ these two linear forces exactly offset, so the slope of the promise objective changes sign and the best reply switches from an interior or central promise to a boundary (more polarized) promise, producing the observed jump; equivalently, expected utility from keeping the promise equals that from reverting to the bliss point precisely at $a = 2(1 - \gamma)$. A second discrete change can occur at $a = \frac{2\gamma\delta}{2 - \gamma\delta}$, where the interior dishonest promise appears or disappears; crossing this cutover replaces the interior solution by a boundary promise (or vice versa), yielding another jump. The absolute-value spatial term is affine on the relevant interval, so the discontinuities come from these boundary switches driven by the changing weight of the electoral term, not from kinks in the spatial utility itself.

3.4 Non-symmetric bliss points

I now relax symmetry and let candidates have different ideal policies. I focus on the case $x_R > 1$ and $x_L = -1$, so the right candidate is farther from the voter than her opponent. I take the feasible policy set to be $[-1, x_R]$ and keep the voting rule unchanged. The last-period logic is unaffected, except that if R wins in period 1 she implements x_R rather than 1.

With asymmetric bliss points, the winner's period-0 implementation rule becomes

$$\pi_R^0 = \begin{cases} p_R^0 & \text{if } a < \gamma\delta \text{ and } p_R^0 \geq x_R - \gamma, \\ x_R & \text{otherwise,} \end{cases} \quad \pi_L^0 = \begin{cases} p_L^0 & \text{if } a < \gamma\delta \text{ and } p_L^0 \leq -1 + \gamma, \\ -1 & \text{otherwise.} \end{cases}$$

Relative to the symmetric case, when R is dishonest she reverts to x_R (her bliss), not to 1. The honesty bands are asymmetric: it takes promises farther from zero to discipline the more distant candidate R (the threshold $x_R - \gamma$), while the left candidate's honesty band stays anchored at $-1 + \gamma$. The analysis is reversed if $x_R < 1$. This means that more radical candidates are less likely to be honest in equilibrium under the same platforms.

Now, I show that there exists an equilibrium where one candidate is honest if she wins while the other is dishonest just from the asymmetry of preferred policies. Because symmetry no longer pins down identical behavior, an equilibrium can feature one honest and one dishonest candidate. I focus on the configuration where L is honest on path and R reneges. Write $(\mathbf{p}, \boldsymbol{\pi}) = (p_L^0, p_R^0, \pi_L^0, \pi_R^0)$. The period-0 expected payoffs in the two win-probability regimes are

$$\mathbb{E}[U_L(\mathbf{p}, \boldsymbol{\pi})] = \begin{cases} -\frac{1}{2a}(a - p_R^0 - p_L^0)(1 + x_R) - \frac{1}{2a}(a + p_R^0 + p_L^0)(1 + p_L^0 + \gamma) & \text{if } \delta|x_R - p_R^0| > a, \\ -\frac{1}{2a}(a - p_R^0 - p_L^0)\left(1 + x_R + \frac{\gamma}{a}(a - \delta(x_R - p_R^0))\right) - \frac{1}{2a}(a + p_R^0 + p_L^0)(1 + p_L^0 + \gamma) & \text{if } \delta|x_R - p_R^0| < a, \end{cases}$$

$$\mathbb{E}[U_R(\mathbf{p}, \boldsymbol{\pi})] = \begin{cases} -\frac{1}{2a}(a - p_R^0 - p_L^0)(2\gamma) - \frac{1}{2a}(a + p_R^0 + p_L^0)(x_R - p_L^0 + \gamma) & \text{if } \delta|x_R - p_R^0| > a, \\ -\frac{1}{2a}(a - p_R^0 - p_L^0)\frac{\gamma}{a}(a + \delta(x_R - p_R^0)) - \frac{1}{2a}(a + p_R^0 + p_L^0)(x_R - p_L^0 + \gamma) & \text{if } \delta|x_R - p_R^0| < a. \end{cases}$$

In the saturated region $\delta|x_R - p_R^0| > a$ the right candidate's payoff is maximized at $p_R^0 = 0$. Hence, whenever $\delta x_R > a$ there is a unique best reply $p_R^0 = 0$ for R , and L best responds by choosing p_L^0 to maximize $\mathbb{E}[U_L]$. Solving the unconstrained problem yields

$$p_L^0 = \frac{x_R - \gamma - p_R^0 - a}{2}.$$

Intersecting best replies gives the subgame pair $(p_R^0, p_L^0) = (0, (x_R - \gamma - a)/2)$. Two checks are needed. First, feasibility for L requires $p_L^0 \in [-1, 0]$, which is satisfied under the parameter range below. Second, L must be honest on path, which holds if $p_L^0 \leq -1 + \gamma$. This inequality is equivalent to

$$\frac{x_R - \gamma - a}{2} < -1 + \gamma \iff 2 + x_R < a + 3\gamma.$$

Proposition 3.5 (Asymmetric honesty under $x_R > 1$). *Suppose $x_R > 1$, $a < \gamma\delta$, $\delta x_R > a$, and*

$$2(x_R - \gamma) > a > 2 + x_R - 3\gamma.$$

Then there exists a period-0 equilibrium in which the left candidate announces and keeps $p_L^0 = \frac{x_R - \gamma - a}{2} \in [-1, 0]$, while the right candidate announces $p_R^0 = 0$ and reneges, implementing $\pi_R^0 = x_R$. By symmetry of the argument, if instead $x_R < 1$ (the right candidate is closer and the left is farther), an analogous equilibrium exists with roles reversed.

Intuitively, when the right candidate is sufficiently distant and reputational discipline bites in period 1, R prefers to center her promise to maximize immediate competitiveness and then revert to x_R in office, while L can profitably commit to a modestly polarized promise that she will keep. The sufficient condition $a < 2(x_R - \gamma)$ tilts R toward shirking, and the honesty band $2 + x_R < a + 3\gamma$ ensures L 's promise lies within her keep-promise region. Taken together, these conditions deliver an asymmetric honesty outcome in which the more distant candidate lies and the closer candidate is disciplined to tell the truth.

3.5 Voter welfare and policy implications

I now evaluate voter welfare across the three equilibrium regions in Theorem 3.1, using three complementary lenses: a utilitarian view based on implemented policy, an honesty and institutional trust perspective based on promise keeping, and a polarization perspective based on platform polarization.

Utilitarian efficiency. If all we cared about was how close the implemented policy is to the voter's ideal, the honest region is the only one that helps: when honesty is sustained, the period 0 policy sits at a moderate distance from the center that scales with uncertainty, while period 1 still delivers an extreme policy. In the other regions the winner implements a bliss point already in period 0, so both periods are extreme. In short: honesty improves the *first* period by moving the outcome toward the center, but nothing fixes the *last* period in this two election environment. This means social welfare is suboptimal in all equilibrium scenarios we identified. Ideally, if candidates could credibly commit, they would converge to moderate policies that maximize overall voter satisfaction (as classic theory predicts). But with only retrospective enforcement, this model shows the system fails to deliver that welfare-maximizing outcome. Voters collectively would be better off if moderate promises could be trusted and implemented – yet such an equilibrium cannot arise under the current rules of the game, which is a troubling inefficiency from a societal standpoint.

Honesty and institutional trust. Another important dimension is the honesty of candidates and the resulting trust in institutions. Voters value honesty for its own sake and

because it underpins the legitimacy of democracy. Equilibria with promise-breaking undermine this: politicians say one thing to get elected and do something else afterward. Such behavior can breed cynicism and distrust among the electorate. Voters may start to view all campaign promises as cheap talk, eroding the accountability that elections are supposed to ensure.

On the other hand, when candidates do keep promises in this model, those promises tend to be extreme policies that many voters dislike – which means even though the politician is honest, a large portion of voters is unhappy with the outcome. This presents a painful trade-off: either voters get deceived (hurting trust in government) in exchange for the hope of a better policy, or they get honesty at the cost of a policy that makes them worse off. Neither scenario is healthy for a democracy. The former leads to disillusionment and skepticism about politicians (*why believe campaign platforms if the winners routinely break their word?*), while the latter can lead to frustration that the political process yields policies misaligned with majority preferences. In short, the results highlight a democratic dilemma: ensuring honesty may come at the cost of policy welfare, and chasing good policies via campaign rhetoric may come at the cost of honesty. This insight underscores why readers should care – it speaks to ongoing concerns about declining trust in political institutions and the prevalence of “political lies” in elections.

Polarization of platforms and outcomes. A core question in electoral politics is whether competition drives candidates toward the center (to appeal to the median voter) or toward the extremes (due to ideology or base mobilization). Classic models with binding commitments predict platform centering – candidates should promise the moderate policy that most voters prefer, leading to convergence at the middle. Our model, by contrast, helps explain why we often observe polarization instead of convergence. Because promises are not binding and voters are backward-looking, a moderate promise is not credible unless the candidate truly intends to enact it. An extreme candidate may reason that a centrist pledge made only to secure office will trigger a retrospective penalty when it goes undelivered, so she is better off signaling her genuinely extreme intentions from the start. Meanwhile, a candidate who does promise the center knowing they will break that promise is banking on fooling voters for short-term gain, but this only works until voters catch on and retaliate. Thus, either candidates avoid moderate promises (leading to honest but polarized campaigns), or they converge in rhetoric but not in reality (leading to disillusionment and, eventually, polarized outcomes anyway).

The net effect is that true centrist outcomes are elusive, and this model provides a mechanism for why political competition might remain polarized. This has direct relevance to real-world trends of polarization: it suggests that one reason we don’t see more centrist

governance, even when many voters are centrists, is the credibility problem – moderate platforms can win elections but may not survive the post-election incentives. Over time, this could even encourage more extreme posturing and skepticism about moderate messaging, reinforcing polarization in a vicious cycle. Understanding this dynamic is important for anyone concerned with why our political discourse has become so divided and why the middle ground often falls out of reach.

Given these welfare considerations, it is natural to ask if there are policy changes or cultural shifts that could lead to better outcomes. While a full analysis is beyond the current scope of this paper, its results hint at some possibilities. One idea is improving mechanisms for accountability and commitment: for example, institutional reforms that make campaign promises more binding or at least more transparent might dissuade candidates from making false promises in the first place. If breaking a promise carried a more immediate cost (legal, reputational, or electoral), candidates would have stronger incentives to either stick to moderate pledges or refrain from over-promising. However, the *elimination* of a candidate after a lie could produce a *last-stage effect* where candidates propose straight lies and apply their preferred policies. Another avenue is voter education and forward-looking behavior. If voters become more prospective – evaluating candidates by expected future actions rather than only past lapses – they might be harder to fool with empty promises. Forward-looking voters would scrutinize the realism of each candidate’s platform and their history of consistency before casting a vote, rather than simply punishing broken promises after the fact. This could encourage candidates to align their promises with what they truly intend to do (making moderate promises only if they genuinely plan moderate policies). In effect, a culture of skepticism toward unrealistic promises and reward for credible honesty could mitigate the negative equilibria the model identifies. However, implementing such changes is challenging. The model’s take-home lesson is that the current retrospective voting dynamic contains an inherent tension that won’t vanish easily: as long as politicians can gain by saying one thing and doing another, and voters only hold them accountable after the betrayal, we are likely to see a cycle of either dishonesty or policy extremism. Recognizing this is a first step toward devising better institutions or norms to foster both honesty and socially optimal policy choices.

4 Conclusion

In this paper, I develop a theoretical model to examine the polarization of politicians as a function of uncertainty regarding voter preferences. The model offers clear predictions: integrity influences campaign promises when its salience is sufficiently high and uncertainty is moderate, and the relationship between uncertainty and promises is non-monotonic. When

uncertainty is moderate, candidates diverge from the median voter toward their own preferred policies; when uncertainty becomes too high, the expected integrity cost loses bite and candidates revert to more centered platforms. Across regimes, the implemented policy in the last period is a bliss point, so discipline operates only through the first-period choice and its reputational carryover.

Ultimately, the most valuable lesson that we learn from this paper is that electoral accountability is a double-edged sword. With non-binding promises and retrospective punishment, voters cannot simultaneously secure moderate policies and honesty. Holding leaders accountable for broken promises is essential, but when voters focus solely on retrospective punishment, it may discourage beneficial moderation and reward only those who either lie convincingly or never moderate their stances at all. This insight is both normative and practical: it urges scholars and policymakers to re-think how we might encourage political actors to be honest and adopt centrist policies. By understanding the strategic forces at play, as illuminated by the model, we can better grasp why political outcomes often disappoint voter expectations and what changes might be needed to improve democratic welfare.

The model helps explain persistent polarization. Because moderate promises are not credible unless they will be implemented, candidates either avoid centrist pledges or make them and later break them. Over time, this pushes rhetoric and outcomes away from the center and erodes institutional trust. The analysis points to complementary levers: raising the salience of integrity, improving information to reduce uncertainty, and strengthening commitment or transparency around pledges so that moderate promises are both credible and enforceable. Cultural shifts toward more forward-looking voter evaluation can also mitigate the worst equilibria by rewarding credible moderation rather than ex post punishment alone.

The findings complement work on electoral rhetoric and backward-looking behavior. From a theoretical standpoint, a natural next step is to study an infinitely repeated environment to examine how integrity evolves without last-period effects, and to extend the model to multiple policy dimensions to explore how politicians might strategically lie on secondary issues while keeping promises on salient ones.

The framework suggests two empirical avenues to measure the role of integrity in policy selection. First, a laboratory experiment in which participants alternate as politicians and voters can track how promise–policy gaps shape reelection and whether integrity salience and uncertainty reproduce the equilibrium taxonomy. Second, a field approach combining measures of promise fulfillment with survey evidence on voter reactions can relate the precision of information available to politicians to campaign positioning and flip-flopping.⁸

⁸Websites such as politifact.com track promise fulfillment and can be paired with survey data on voter

Together, these steps would test the model's central claim: that improving information and credible commitment expands the set of states in which honesty is sustained without exacerbating polarization, aligning democratic accountability with better policy.

behavior.

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A Concave vs convex reputation costs

This appendix generalizes the linear integrity cost to a power specification and studies how curvature affects behavior. I keep all primitives of the environment unchanged and let the period-1 penalty be generated by the period-0 promise–policy gap via a generic loss $\mathfrak{C}_i^1 = \delta|p_i^0 - \pi_i^0|^\kappa$ with $\kappa > 0$. Because the penalty is predetermined by period-0 choices, all last-period results are unaffected; the only change concerns the winner’s period-0 policy choice and the induced reputational term carried into period 1. I first provide a formal derivation of the period-0 best reply (a generalization of Proposition 3.3) for the cases $\kappa \in (0, 1)$ and $\kappa \in (1, \infty)$. I then offer an intuitive discussion of how the cost curvature shifts the boundaries in the promises subgame (Proposition 3.4) and the equilibrium taxonomy (Theorem 3.1)—highlighting when the winner’s policy choice is “bang–bang” (keep the promise or revert to the bliss point) versus interior “shading” toward the bliss point.

Think of $\delta|p_i^0 - \pi_i^0|^\kappa$ as how voters convert “broken-promise distance” into an integrity hit. If $0 < \kappa < 1$, the cost is *concave*: the first bit of lying hurts the most, and each additional bit hurts less. Voters are hyper-sensitive to even small breaches, so the winner either keeps the promise when discipline bites or, if the trade-off favors policy payoffs, goes all the way back to her bliss point—hence the *bang–bang* behavior also seen under $\kappa = 1$. If $\kappa > 1$, the cost is *convex*: small deviations are cheap, but the marginal penalty accelerates as the lie grows. Voters are forgiving of tiny adjustments but harsh on big betrayals, so the winner applies policies partway toward her bliss point and stops before large reputational losses kick in. The next two propositions develop this logic formally.

Proposition A.1 ($\kappa < 1$). *If candidate R wins at $t = 0$ with p_R^0 , she chooses in equilibrium*

$$\pi_R^0 = \begin{cases} p_R^0 & \text{if } a \leq \gamma\delta(1 - p_R^0)^{\kappa-1} \text{ and } p_R^0 \geq 1 - \gamma, \\ 1 & \text{otherwise.} \end{cases}$$

See [proof](#) on page 39.

Proposition A.2 ($\kappa > 1$). *If candidate R wins at $t = 0$ with p_R^0 , she chooses in equilibrium*

$$\pi_R^0 = \min \left\{ 1, p_R^0 + \left(\frac{a}{\gamma\delta\kappa} \right)^{\frac{1}{\kappa-1}} \right\}.$$

See [proof](#) on page 39.

When the cost function is *concave* ($0 < \kappa < 1$) and uncertainty is sufficiently low, honesty remains feasible; the winner’s optimal choice is at a corner: either $\pi_R^0 = p_R^0$ (keep

the promise) or $\pi_R^0 = 1$ (revert to the bliss point), as in the linear case $\kappa = 1$. When the cost function is *convex* ($\kappa > 1$), honesty does not arise; instead, the winner typically chooses an interior policy $\pi_R^0 \in [p_R^0, 1)$, shading toward her bliss point (with $\pi_R^0 = 1$ only if the constraint binds), so equilibrium behavior is not fully selfish. Notice that a sufficient condition for candidate to apply its bliss point is that $a \geq \delta\gamma\kappa$, which coincides with Proposition 3.3 $a > \gamma\delta$.

A.1 $\kappa < 1$

When the integrity cost is concave ($\kappa < 1$), the qualitative structure of Proposition 3.4 is preserved: equilibrium promises still fall into the same three regions as in the linear benchmark, but the cutoffs that separate them shift. In particular, the low-uncertainty boundary $a < \gamma\delta$ is replaced by a threshold that depends on the promised distance to the winner's bliss point, scaling with $(1 - p_R^0)^\kappa$ (reflecting the power loss). Thus, the same three families of promises are selected in equilibrium, with boundaries re-indexed by κ . The main difference is that, in the high-uncertainty region (the analogue of $a > \gamma\delta$ in Proposition 3.4), the interior promise that induces dishonesty no longer admits a closed-form expression; it is characterized implicitly by the first-order condition equating marginal electoral gains and marginal reputational losses, and must be obtained numerically. Moreover, the existence of this interior optimizer depends on the concavity or convexity of the expected utility function that does not only depend on the κ parameter.

The partition in Theorem 3.1 survives: there remain three qualitatively distinct regions (an honest region at low uncertainty, a polarized but honest region at intermediate parameters, and a dishonest region at higher uncertainty or lower integrity salience), but the cutoffs shift as functions of κ and of the promised distance to the bliss point. Because the first unit of deviation is heavily punished, the winner's period 0 policy choice is bang-bang (keep the promise or jump to the bliss point), and promises are chosen to balance that discontinuous response. The non-monotone pattern in promises as uncertainty varies is preserved, the honest wedge at low uncertainty tends to expand relative to the linear case, and interior dishonest promises in the high-uncertainty region are typically characterized implicitly rather than in closed form.

A.2 $\kappa > 1$

When the integrity cost is convex ($\kappa > 1$), the honest region in Proposition 3.4 disappears: along the equilibrium path the period 0 winner deviates from her promise by an interior

amount. The same three qualitative families of promises remain relevant, but the cutoffs shift, with the low-uncertainty boundary $a < \gamma\delta$ replaced by a threshold that depends on the promised distance to the winner's bliss point through a positive power of $(1 - p_R^0)$. On the region where the reputational term enters explicitly ($\delta|p_R^0 - \pi_R^0|^\kappa < a$), expected utility is the sum of an affine term and a negative convex loss, hence it is concave in the implemented policy; on the saturated region it is affine. Anticipating the interior shading and piecewise concavity, candidates choose period 0 promises closer to their bliss points to cap the future promise–policy gap, so equilibrium promises are more polarized than in the linear benchmark. As in the concave case, the interior dishonest promise in the high-uncertainty region does not admit a closed-form expression and is defined implicitly by the first-order condition balancing marginal electoral gains and marginal reputational losses.

Theorem 3.1 loses the honest region: the marginal penalty at zero is weak, so the period 0 winner always shades away from the promise by an interior amount. The same families of promises (central versus polarized) still organize the equilibrium, but the cutoffs move, and—anticipating interior shading—candidates select more polarized promises (closer to their bliss points) to cap the future promise–policy gap. The additive-valence interpretation of second-period discipline remains, though the region in which a deviator is certain to lose the next election can shrink because optimal shading dampens the reputational hit. Overall, the comparative statics in integrity salience and patience continue to discipline behavior, while convexity tilts the taxonomy toward polarization and universal post-election deviation.

B Mathematical appendix

Proof of Proposition 3.1. The last period flow utility of candidate i if she wins at $t = 1$ is

$$u_i(\pi_i^1; p_i^1, p_{-i}^1, \pi_{-i}^1) = -(|x_i - \pi_i^1|)$$

Notice that this utility function only depends on π_i^1 , a variable that the candidate chooses, and x_i which is exogenous. Then, this expression is maximized at $\pi_i^1 = x_i$ for every pair p_i^1, p_{-i}^1 . $\square$

Proof of Proposition 3.2. The expected utility in equilibrium for candidate R given p_L^1 at period 1 is:

$$\mathbb{E}(u_R^1(p_R^1; p_L^1, \pi_L^1, \pi_R^1)) = \begin{cases} -\frac{1}{a}(a + |p_R^1| - |p_L^1| - \mathfrak{C}_L^1) & \text{if } a + |p_R^1| - |p_L^1| - \mathfrak{C}_L^1 \in [-a, a] \\ 0 & \text{if } a + |p_R^1| - |p_L^1| - \mathfrak{C}_L^1 > a \\ -2 & \text{if } a + |p_R^1| - |p_L^1| - \mathfrak{C}_L^1 < -a \end{cases}$$

If $\mathfrak{C}_L \geq a$, for every p_L^1 , there is a p_R^1 close enough to zero such that candidate R wins with probability one and gets zero utility at $t = 1$. In particular, given p_L^1 , candidate R 's best reply must meet the following condition

$$|p_R^1| \leq |p_L^1| + \mathfrak{C}_L^1 - a$$

If candidate L plays a promise different from zero in equilibrium, candidate R can best reply with a promise bigger, in absolute value, than $\mathfrak{C}_L^1 - a$. Notice that this can not happen in equilibrium, as candidate L will be willing to deviate and choose a p_L^1 closer to zero such that the previous condition fails. Hence, for $\mathfrak{C}_L^1 > a$ there are multiple equilibria where candidate R plays $p_R^1 \in [a - \mathfrak{C}_L^1, \mathfrak{C}_L^1 - a]$. As candidate L can not choose any promise to win, she is indifferent to any promise that she can choose. In equilibrium, $p_L^1 \in \mathbb{R}$.

If $a > \mathfrak{C}_L^1 \geq 0$, candidate L can guarantee herself a positive probability of winning choosing $p_L^1 = 0$, as $\mathfrak{C}_L^1 - a < 0$ and $|p_R^1| \leq 0$. Both candidates, choosing a promise close to the voter's ideal policy, can have a strictly positive probability of winning. With this, candidate's R utility maximization is equivalent to maximize

$$-\frac{1}{a}(a + |p_R^1| - |p_L^1| - \mathfrak{C}_L^1)$$

which is decreasing for $|p_R^1|$. Reasoning equivalently for candidate L , I can state that both candidates best reply to each other with $p_R^1 = p_L^1 = 0$. $\square$

Proof of Proposition 3.3. The expected utility of candidate R at $t = 0$ when she wins with p_R^0 reads as

$$\mathbb{E}[U_R(\pi_R^0; p_R^0)] = \begin{cases} -|1 - \pi_R^0| - \frac{\gamma}{a}(a + \delta|\pi_R^0 - p_R^0|) & \text{if } \delta|\pi_R^0 - p_R^0| < a \\ -|1 - \pi_R^0| - 2\gamma & \text{if } \delta|\pi_R^0 - p_R^0| > a \end{cases} \quad (7)$$

The two parts of the function depend on the integrity cost. Equivalently, I can say that the difference between the policy and the promise determines the boundaries of the utility

function. For both parts, the first term corresponds to the utility derived from applying the policy π_R^0 . The second term is the expected utility in the last period of the game. It's important to note two issues with this function. First, the function is continuous for every π_R^0 . In particular, when $\delta|\pi_R^0 - p_R^0| = a$, as

$$\lim_{\mathfrak{C}_R^0 \rightarrow a^+} \mathbb{E} [U_R(\pi_R^0; p_R^0)] = \lim_{\mathfrak{C}_R^0 \rightarrow a^-} \mathbb{E} [U_R(\pi_R^0; p_R^0)]$$

Second, the function has a unique maximum for $\mathfrak{C}_R^0 > a$, and it is $\pi_R^0 = 1$. Consequently, to find the maximum, it is enough to study the π_R^0 maximizing the function when $\mathfrak{C}_R^0 \leq a$ and comparing the values of the expected utility.

The expected utility when $\mathfrak{C}_R^0 \leq a$ is increasing on $\pi_R^0 \in [p_R^0, 1]$ if $1 > \frac{\gamma\delta}{a}$. In this case, $\pi_R^0 = 1$ is the maximum of this function.

If $1 < \frac{\gamma\delta}{a}$, the function is decreasing in the same interval for π_R^0 . Hence, the maximum of the function will be p_R^0 if the first part of the function is the biggest, and π_R^0 otherwise:

$$\pi_R^0 = \begin{cases} p_R^0 & \text{if } -|1 - p_R^0| - \gamma \geq -2\gamma \\ 1 & \text{if } -|1 - p_R^0| - \gamma < -2\gamma \end{cases}$$

So p_R^0 is the maximum if

$$|1 - p_R^0| \leq \gamma$$

Rearranging terms, we have

$$1 + \gamma \geq p_R^0 \geq 1 - \gamma$$

Therefore,

$$\pi_R^0 = \begin{cases} p_R^0 & \text{if } a < \gamma\delta \text{ and } 1 + \gamma \geq p_R^0 \geq 1 - \gamma \\ 1 & \text{Otherwise} \end{cases}$$

□

Proof of Proposition 3.4. If $a < \gamma\delta$ and $p_R^0 \geq (1 - \gamma)$, Proposition 3.3 states that the candidates will apply their promise if they win. The expected utility for candidate R is then,

$$\begin{aligned}\mathbb{E}[U_R(\mathbf{p}, \boldsymbol{\pi})] &= -\frac{1}{2a} (-|p_R^0| + |p_L^0| + a) [1 - p_R^0 + \gamma] \\ &\quad -\frac{1}{2a} (a + |p_R^0| - |p_L^0|) [|1 - p_L^0| + \gamma]\end{aligned}$$

Notice that $|p_R^0| > 1$ can not happen in equilibrium, as $p_R^0 = 1$ gives higher utility when winning and increases or keeps the probability of winning. In equilibrium, it must be that $p_R^0 \geq 0$, as for any promise smaller than zero, promising the policy of the voter increases the probability and the utility of winning. Reasoning similarly for candidate L , the expected utility can be rewritten as

$$\begin{aligned}\mathbb{E}[U_R(\mathbf{p}, \boldsymbol{\pi})] &= -\frac{1}{2a} (-p_R^0 - p_L^0 + a) [1 - p_R^0 + \gamma] \\ &\quad -\frac{1}{2a} (a + p_R^0 + p_L^0) [1 - p_L^0 + \gamma]\end{aligned}$$

Solving the utility maximization problem for p_R^0 and doing the same for candidate L , I get the best replies of each candidate.

$$p_R^0 = \begin{cases} \frac{a}{2} & \text{if } a < 2 \\ 1 & \text{if } a \geq 2 \end{cases}, \quad p_L^0 = \begin{cases} -\frac{a}{2} & \text{if } a < 2 \\ -1 & \text{if } a \geq 2 \end{cases}$$

But notice that if $a \geq 2$,

$$a \geq \frac{a}{2} \geq 1 \geq \gamma\delta$$

This statement is false by assumption whenever $\delta \leq 1$. However, it can be true when $\delta > 1$. Therefore, if $\gamma\delta > a$ and $a \geq 2$, $p_R^0 = 1 = -p_L^0$ and, in any other case, $p_R^0 = \frac{a}{2}$. Therefore, if $a < \gamma\delta$ and $a \geq 2(1 - \gamma)$

$$p_R^0 = \frac{a}{2}, \quad p_L^0 = -\frac{a}{2}$$

Assume now that $a < \gamma\delta$ and $a < 2(1 - \gamma)$. Candidate R will only apply her promise if $p_R^0 > 1 - \gamma$. If not, the expected utility of candidate R as a function of p_R^0 is

$$\begin{aligned} \mathbb{E}[U_R(\mathbf{p}, \boldsymbol{\pi})] = & -\frac{1}{2a} (-|p_R^0| + |p_L^0| + a) \left[\frac{\gamma}{a} (\delta|1 - p_R^0| + a) \right] \\ & - \frac{1}{2a} (a + |p_R^0| - |p_L^0|) \left[2 + \frac{\gamma}{a} (-\delta|1 - p_R^0| + a) \right] \end{aligned} \quad (8)$$

when $\delta|1 - p_R^0| < a$, and

$$\mathbb{E}[U_R(\mathbf{p}, \boldsymbol{\pi})] = -\frac{1}{a} (-|p_R^0| + |p_L^0| + a) \gamma - \frac{1}{a} (a + |p_R^0| - |p_L^0|) \quad (9)$$

if $\delta|1 - p_R^0| > a$.

Notice that the expected utility in (9) reaches a maximum at $p_R^0 = 0$. On the other hand, using the same reasoning of the previous part of the proof, $p_R^0 \in [0, 1]$ and $p_L^0 \in [-1, 0]$. And then, I can write equation (8) as

$$\begin{aligned} \mathbb{E}[U_R(\mathbf{p}, \boldsymbol{\pi})] = & -\frac{1}{2a} (-p_R^0 - p_L^0 + a) \left[\frac{\gamma}{a} (\delta(1 - p_R^0) + a) \right] \\ & - \frac{1}{2a} (a + p_R^0 + p_L^0) \left[2 + \frac{\gamma}{a} (-\delta(1 + p_L^0) + a) \right] \end{aligned}$$

The unconstrained maximization problem of the previous equation gives as a maximum $p_R^* = 1 + \frac{a}{2} - \frac{a}{\gamma\delta}$. By symmetry, we can see that $p_L^0 = -1 - \frac{a}{2} + \frac{a}{\gamma\delta}$.

If $\delta(1 - p_R^0) \geq a$, candidate R chooses in equilibrium $p_R^0 = 0$. This is because the integrity cost in which candidate R incurs given his optimal promise is too high, and therefore she can choose a better promise. The same holds for candidate L . If instead, $\delta(1 - p_R^0) < a$, we have that developing the inequality,

$$1 > \frac{2 - \gamma\delta}{2\gamma} \quad (10)$$

This means that $p_R^* < 0$, as

$$1 \leq \frac{a}{\delta} \left(\frac{2 - \gamma\delta}{2\gamma} \right)$$

where $\frac{a}{\delta} < \gamma \leq 1$ and the right term is smaller than one. Notice that condition (10) can be rewritten as $\gamma\delta > 2(1 - \gamma)$. Therefore, if $\gamma\delta > 2(1 - \gamma) > a$ both candidates choose $p_i^0 = 0$ in equilibrium.

If $a > \gamma\delta$, candidate R will apply the policy $\pi_R^0 = 1$ if she wins. Hence, the expected utility of candidate R reads as

$$\begin{aligned} \mathbb{E}[U_R(\mathbf{p}, \boldsymbol{\pi})] = & -\frac{1}{2a} (-|p_R^0| + |p_L^0| + a) \left[\frac{\gamma}{a} (\delta|1 - p_R^0| + a) \right] \\ & - \frac{1}{2a} (a + |p_R^0| - |p_L^0|) \left[2 + \frac{\gamma}{a} (-\delta|1 - p_L^0| + a) \right] \end{aligned} \quad (11)$$

when $\delta|1 - p_R^0| < a$. Using the same reasoning than previously,

$$\begin{aligned} \mathbb{E}[U_R(\mathbf{p}, \boldsymbol{\pi})] = & -\frac{1}{2a} (-p_R^0 - p_L^0 + a) [\gamma (\delta(1 - p_R^0) + a)] \\ & - \frac{1}{2a} (a + p_R^0 + p_L^0) \left[2 + \frac{\gamma}{a} (-\delta(1 + p_L^0) + a) \right] \end{aligned}$$

If $\delta|1 - p_R^0| > a$, the expected utility is

$$\mathbb{E}[U_R(\mathbf{p}, \boldsymbol{\pi})] = -\frac{1}{a} (-|p_R^0| + |p_L^0| + a) \gamma - \frac{1}{a} (a + |p_R^0| - |p_L^0|)$$

This function, when $\delta|1 - p_R^0| > a$, has as maximum at $p_R^0 = 0$. The unconstrained maximization problem of the expected utility when $\delta|1 - p_R^0| < a$ gives as a maximum $p_R^* = 1 + \frac{a}{2} - \frac{a}{\gamma\delta}$. By symmetry, we can see that $p_L^0 = -1 - \frac{a}{2} + \frac{a}{\gamma\delta}$.

Notice that, as the expected utility function is continuous when $\delta|1 - p_R^0| = a$, I only need to compare the utility of $p_R^0 = 0$ against p_R^* to know which is the maximum. Two conditions must hold for p_R^* being a best reply in equilibrium:

1. $p_R^* \geq 0$. This implies, using the same procedure of the previous proposition,

$$\gamma\delta \geq \frac{2a}{2 + a}$$

2. $\delta(1 - p_R^*) < a$. This implies

$$1 > \frac{2 - \gamma\delta}{2\gamma}$$

If any of these two condition fails, candidate R prefers to choose in equilibrium $p_R^0 = 0$. This can be either because $\delta < a$ and the candidate in equilibrium has a positive probability of winning in the second stage or either because even having zero probability of winning, the candidate values more increasing the probability of winning than the integrity cost.

□

Proof of Theorem 3.1. The proof is done in three different parts, each one corresponding to the three different types of equilibria.

1. If $2(1 - \gamma) < \gamma\delta$ the optimal promises in equilibrium are determined by proposition ??, and the policies by proposition 3.3. As candidates apply their promise, they arrive to the second stage with same level of integrity, and hence, they will play the most moderate policy and chose their ideal policy if they win.
2. In this case, promises are those stated at the beginning of this section. Proposition 3.3 ensure that the winner will apply her ideal policy. The lying costs of the candidate that wins are

$$\mathfrak{C}_i^t = \delta \left| 1 + \frac{a}{2} - \frac{a}{\gamma\delta} - 1 \right| = \delta a \left| \frac{2 - \gamma\delta}{2\gamma\delta} \right|$$

At the beginning of this section I characterize the promises in equilibrium depending on the relation between integrity and uncertainty. By assumption,

$$\delta a \left| \frac{2 - \gamma\delta}{2\gamma\delta} \right| < a$$

or, equivalently,

$$\left| \frac{2 - \gamma\delta}{2\gamma} \right| < 1 \Rightarrow 2(1 - \gamma) < \gamma\delta$$

Therefore, both candidates promise the ideal policy of the median voter and apply their ideal policy if they win.

3. For the third type of equilibrium, either $\gamma\delta < 2(1 - \gamma)$ or $\gamma\delta < \frac{2a}{2+a}$. In the both cases, the optimal promises are determined by proposition ?. Proposition 3.3 guarantees that the winner applies her ideal policy. She incurs on an integrity cost equal to δ . From the characterization of promises, if $\delta < a$ there is a unique equilibrium in which both candidates play $p_i^1 = 0$ and the winner applies her preferred policy. If, instead, $\delta > a$, the advantaged candidate can play anything in the interval $[a - \delta, \delta - a]$ to win and apply her ideal policy with probability one and the other candidate can not play any promise to win.

□

Proof of Proposition A.1. On the region where $\delta|\pi_R^0 - p_R^0|^\kappa < a$,

$$\mathbb{E}[U_R(\pi_R^0, p_R^0)] = -|1 - \pi_R^0| - \frac{\gamma}{a}(a + \delta|\pi_R^0 - p_R^0|^\kappa).$$

Since $\pi_R^0 \in [p_R^0, 1]$, write $x = \pi_R^0 - p_R^0 \in [0, 1 - p_R^0]$ and note $|1 - \pi_R^0| = 1 - p_R^0 - x$. For $0 < \kappa < 1$, x^κ is concave, so $-\frac{\gamma}{a}\delta x^\kappa$ is convex; adding the affine term $-(1 - p_R^0 - x)$ yields a convex objective on this region. Hence, any interior point is a minimum, and the maximum is attained at a boundary. Comparing the two boundaries,

$$\pi_R^0 = 1 \iff -\frac{\gamma}{a}(a + \delta|1 - p_R^0|^\kappa) \geq -|1 - p_R^0| - \gamma,$$

which is equivalent to $\gamma\delta|1 - p_R^0|^{\kappa-1} \leq a$. Finally, comparing payoffs across the two pieces (unsaturated vs. saturated) implies $p_R^0 > 1 - \gamma$ is required for $\pi_R^0 = p_R^0$ to be a best reply. □

Proof of Proposition A.2. For $\kappa > 1$ and on the region $\delta|\pi_R^0 - p_R^0|^\kappa < a$, the term $-\frac{\gamma}{a}\delta|\pi_R^0 - p_R^0|^\kappa$ is concave; adding the affine $-|1 - \pi_R^0|$ preserves concavity, so any interior stationary point is a maximum. Using $x = \pi_R^0 - p_R^0$, the FOC gives $1 - \frac{\gamma}{a}\delta\kappa x^{\kappa-1} = 0$, hence $x^* = \left(\frac{a}{\gamma\delta\kappa}\right)^{\frac{1}{\kappa-1}}$. Imposing the constraint $x \in [0, 1 - p_R^0]$ yields $\pi_R^0 = p_R^0 + \min\{x^*, 1 - p_R^0\}$. If instead $\delta|\pi_R^0 - p_R^0|^\kappa \geq a$, expected utility is linear in π_R^0 and maximized at $\pi_R^0 = 1$. □

Proof. (Promises in the first stage) As shown previously, payoffs in equilibrium do not depend on promises. The expected utility in equilibrium for candidate R given p_L^1 at period 1 is

$$\mathbb{E}(u_R^1(p_R^1; p_L^1, \pi_L^1, \pi_R^1)) = P(\text{Win}_L)(-2) = \begin{cases} \frac{-1}{a}(a + |p_R^1| - |p_L^1| - \mathfrak{C}_L^t) & \text{if } -|p_R^1| + |p_L^1| + \mathfrak{C}_L^t \in [-a, a] \\ 0 & \text{if } -|p_R^1| + |p_L^1| + \mathfrak{C}_L^t > a \\ -2 & \text{if } -|p_R^1| + |p_L^1| + \mathfrak{C}_L^t < -a \end{cases}$$

If $\mathfrak{C}_L \geq a$, for every p_L^1 there is a p_R^1 close enough to zero such that candidate R wins with probability one and gets zero utility at $t = 1$. In particular, given p_L^1 , candidate R best reply must meet the following condition

$$|p_R^1| \leq |p_L^1| + \mathfrak{C}_L^1 - a$$

If candidate L plays a promise different from zero in equilibrium, candidate R can best reply with a promise bigger, in absolute value, than $\mathfrak{C}_L^1 - a$. Notice that this can not happen

in equilibrium, as candidate L will be willing to deviate and choose a p_L^1 closer to zero such that the previous condition fails. Hence, for $\mathfrak{C}_L^1 > a$ there are multiple equilibria where candidate R plays $p_R^1 \in [a - \mathfrak{C}_L^1, \mathfrak{C}_L^1 - a]$. As candidate L can not choose any promise to win, she is indifferent to any promise that she can choose. In equilibrium, $p_L^1 \in \mathbb{R}$.

If $a > \mathfrak{C}_L^1 \geq 0$, candidate L can guarantee herself a positive probability of winning choosing $p_L^1 = 0$, as

$$\mathfrak{C}_L^t - a < 0 \text{ and } 0 \geq |p_R^1|$$

Both candidates, choosing a promise close to the voter's ideal policy, can have a strictly positive probability of winning. With this, candidate's R utility maximization is equivalent to maximize

$$\frac{-1}{a} (b + |p_R^1| - |p_L^1| - \mathfrak{C}_L^t)$$

which is decreasing for $|p_R^1|$. Reasoning equivalently for candidate L , I can state that both candidates best reply to each other with $p_R^1 = p_L^1 = 0$.

□